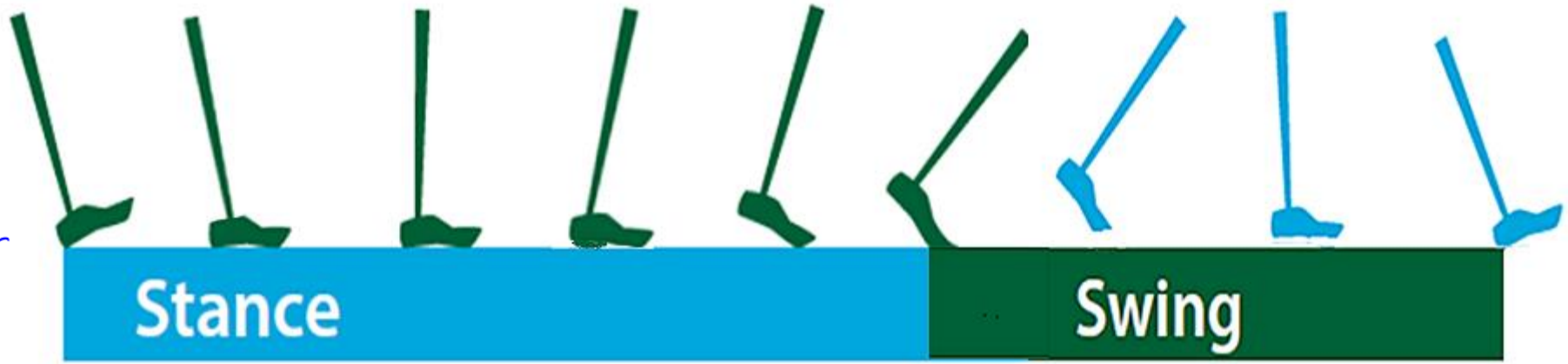
Human gait

One gait cycle – initial contact to initial contact of the same foot



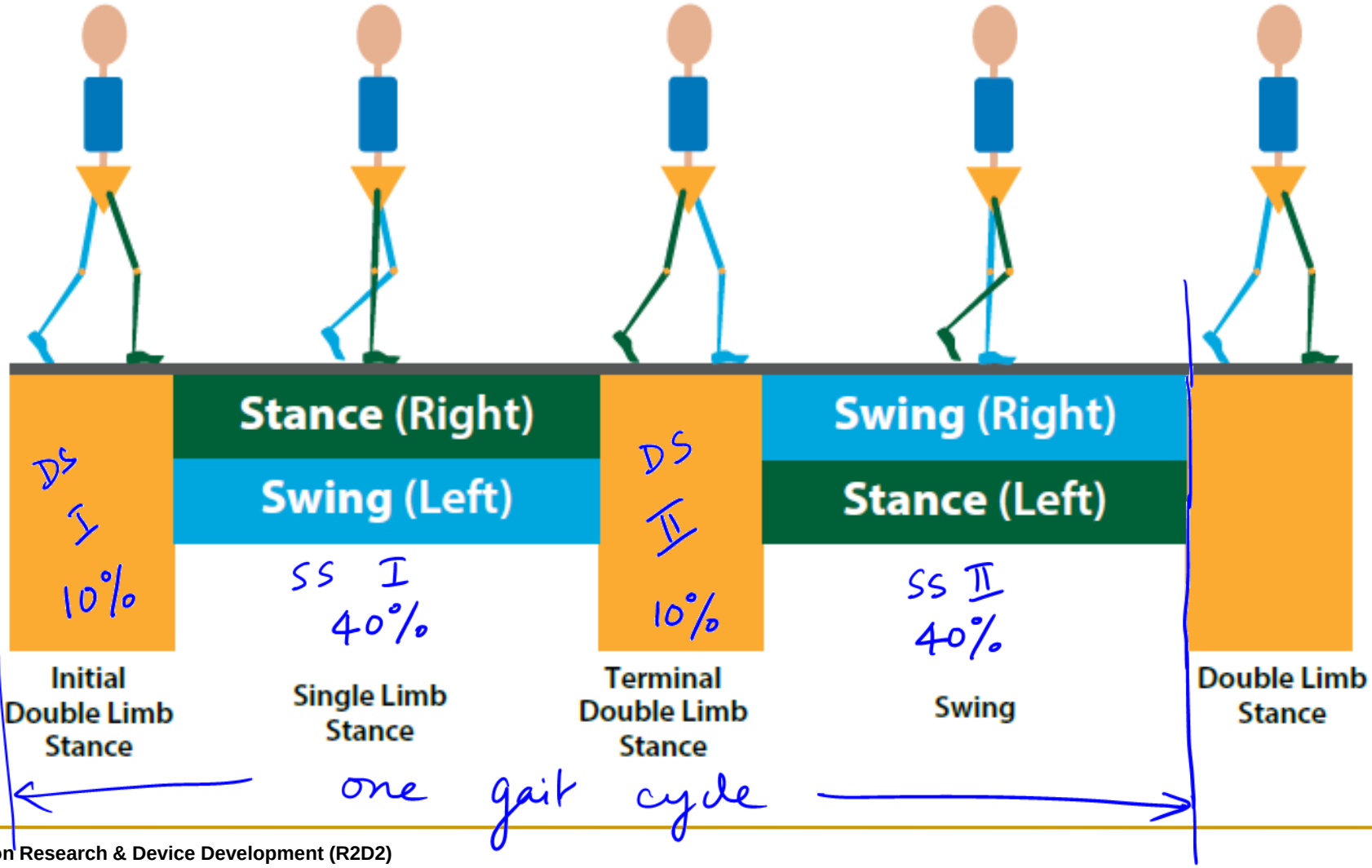
Heel strike
Heel contact
Initial contact

IC & lifting off the foot from the ground – stance

Foot lifted off the ground to "initial" contact of that foot – swing phase

Gait cycle

Both limbs in contact with the ground – Double support phase
When either limb is in contact with the ground – Single support phase

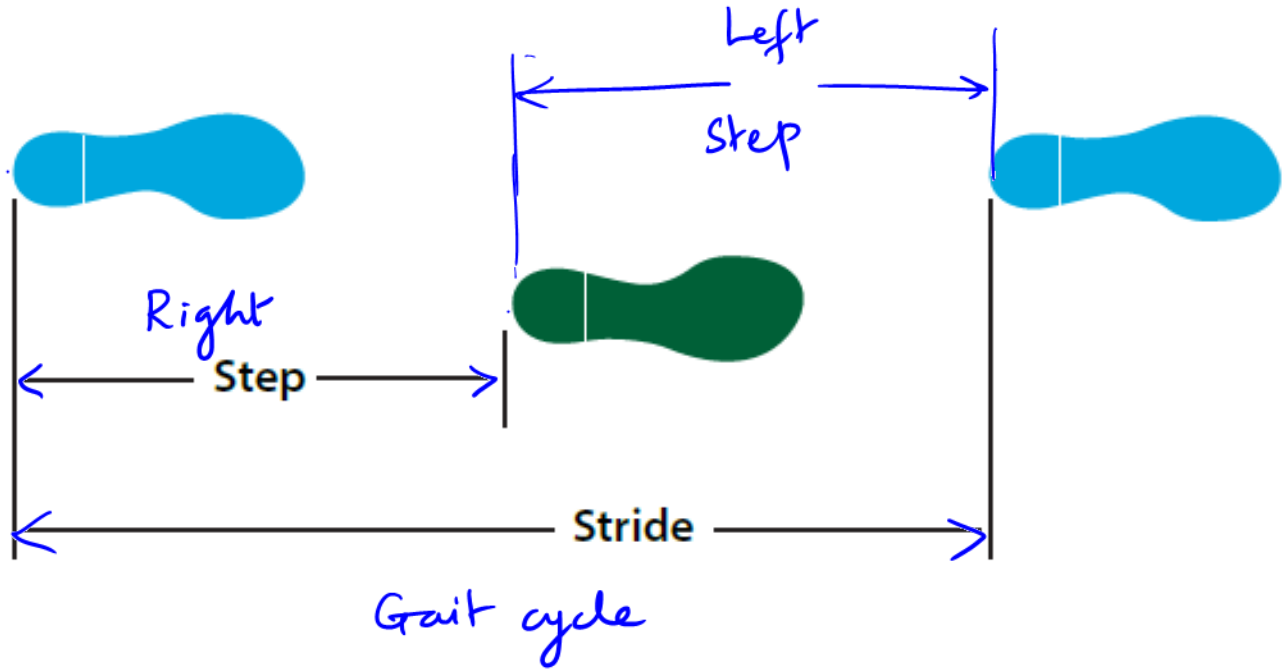


Each limb is in
Stance – 60%
Swing – 40%

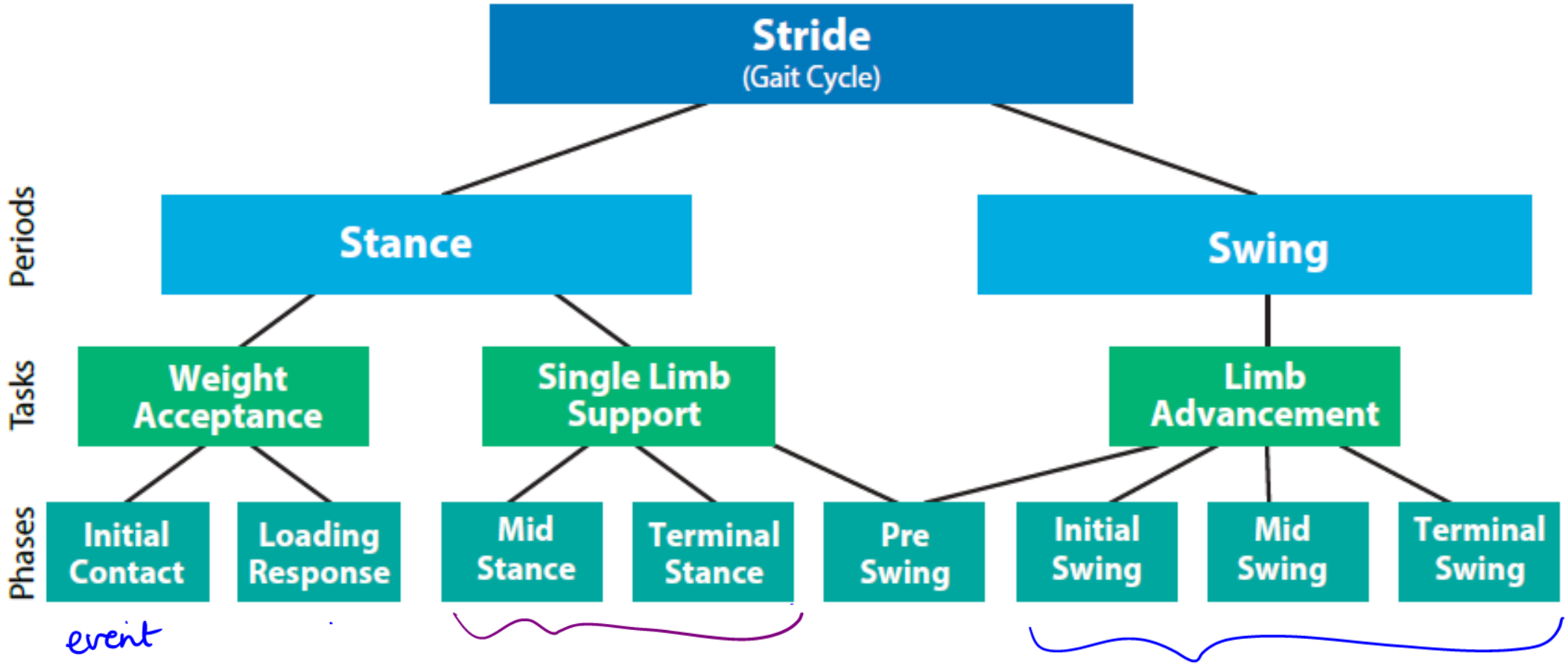
Gait cycle

Normal gait cycle

Two symmetric steps

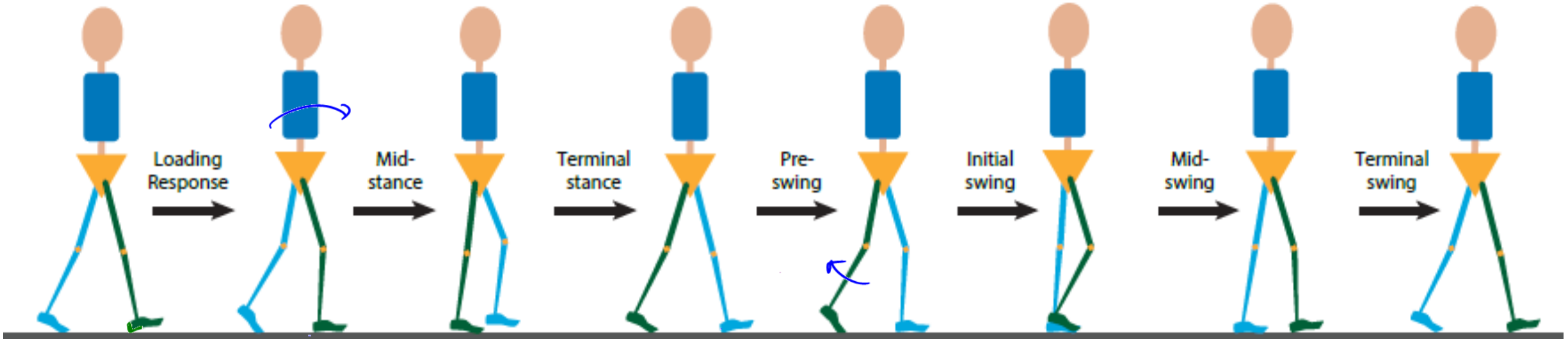


Gait cycle


Stride
(Gait Cycle)
Periods
Stance
Swing
Tasks
Weight Acceptance
Single Limb Support
Limb Advancement
Phases
Initial Contact
Loading Response
Mid Stance
Terminal Stance
Pre Swing
Initial Swing
Mid Swing
Terminal Swing
event

Gait cycle

IC
Initial contact – the moment the foot strikes the ground
Opposite Toe-Off (OTO) – the moment the contralateral foot lifts off the ground
Heel rise – heel of the stance leg lifts off the ground



IC ipsilateral
contralateral

OTO

HR

OIC

TO

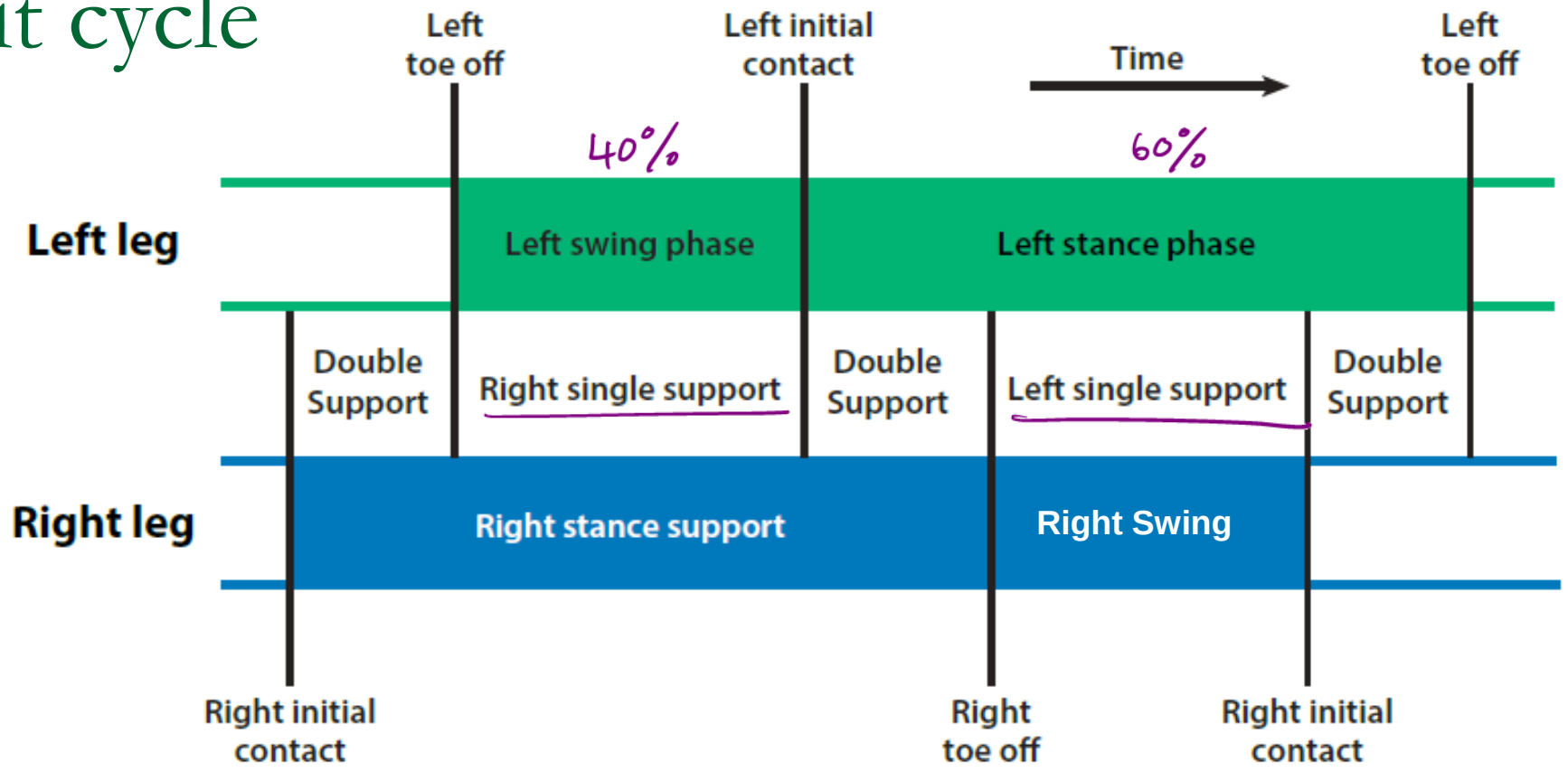
FA

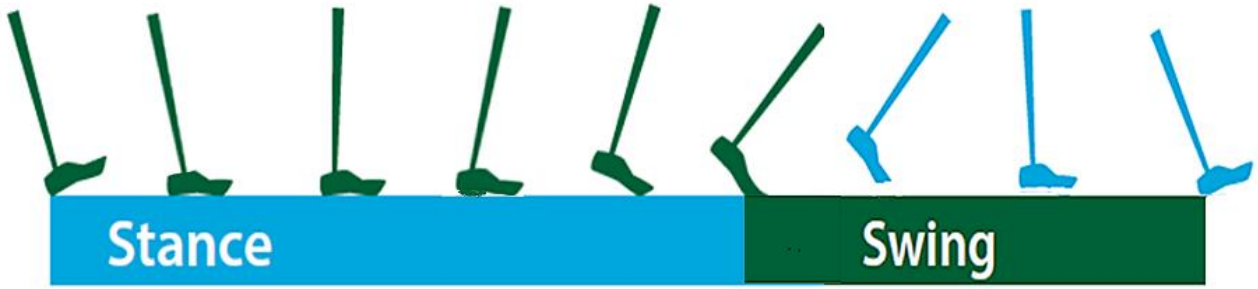
TV

IC for next gait cycle

Foot flat happens before OTO

Gait cycle


Left toe off
Left initial contact
Time
Left toe off
40%
60%
Left leg
Left swing phase
Left stance phase
Double Support
Right single support
Double Support
Left single support
Double Support
Right leg
Right stance support
Right Swing
Right initial contact
Right toe off
Right initial contact


Stance
Swing

Initial contact (IC) : The moment the foot strikes the ground

Loading response (LR) : Initial DS period when the limb is accepting weight

Mid-stance : First phase of SS - body advances over the supporting limb and moves ahead of the stance limb as weight is transferred to the forefoot

Terminal stance : Last phase of SS & it ends with OIC

Pre-swing : Final DS period in preparation for swing and weight is shifted to the opposite limb.

Preswing of one limb corresponds to LR of the other limb

Initial swing : First third of swing period - ends with feet adjacent

Mid-swing : Middle third of swing period, ends with a vertical tibia

Terminal-swing : Last third of swing - knee extends to position for next IC